



Dolivo Overview & Case Studies

Specialist power systems due diligence for utility-scale solar and battery assets

Dolivo helps renewable energy developers, IPPs, asset owners and investors access specialist power systems engineering across the project lifecycle. We support technical decisions around grid connection, grid compliance, electrical design, plant controls, commissioning readiness and operational performance.

Our work combines engineer-led analysis with internal automation tools for reviewing technical documents, processing project data and running repeatable power system assessments. The aim is to make specialist grid and power systems expertise more accessible at the points where it can materially affect project value, deliverability and performance.

Where we support

Development and acquisition	Reviewing grid connection requirements, electrical assumptions and capacity risks before commercial commitments are made.
Detailed design and pre-construction	Assessing whether connection requirements, consultant studies, equipment selections, plant controller requirements and EPC design decisions remain aligned as the project develops.
Commissioning and handover	Reviewing models, plant settings, control-system behaviour and unresolved technical risks before testing and energisation.
Operations and asset management	Analysing operational data to identify grid compliance issues, control-system problems and electrical performance risks, then quantifying where these may lead to fines, lost generation or underperformance and recommending practical fixes.

Core areas of work

- Technical due diligence:** Independent review of grid connection requirements, electrical design assumptions, control systems and material technical risks.
- Grid compliance and model alignment:** Reviewing whether project models, equipment selections and site settings remain consistent with the latest design and grid requirements.
- Control-system and site optimisation:** Assessing plant controller behaviour, inverter settings and transformer configuration to identify compliance or performance improvements.
- Operational diagnostics:** Using SCADA data, PPC configuration information and power system models to identify root causes of grid-related charges, lost generation and unusual electrical behaviour.
- Engineering automation:** Developing internal tools and platform to make technical document review, project data analysis and repeatable power system assessments faster, more consistent and easier to apply across portfolios.

Case Study 1: Grid-related cost exposure on two 100 MW operational solar assets

Customer: Renewable asset owner operating two large utility-scale solar farms

Challenge

The sites were exposed to multi-six-figure annualised grid-related charges linked to import capacity and reactive power behaviour. The customer needed to understand what was driving the charges, whether the issue could be addressed through plant controls, and whether the contracted import capacity remained appropriate for the actual electrical behaviour of the sites.

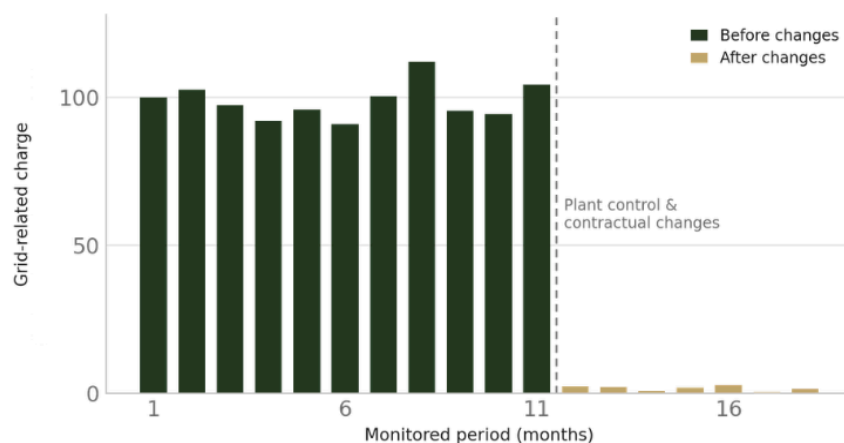
Dolivo's work

Dolivo built an analysis pipeline using SCADA data and power system models to assess site import and reactive power behaviour, night-time operation, transformer energisation, plant controls and contracted import capacity.

The analysis distinguished short-duration events from the sustained operating behaviour responsible for the majority of the exposure.

Recommendations

- Revised import capacity requirements based on actual site behaviour.
- Control-system changes to reduce overnight reactive load.
- Q-at-night strategy with inverter rotation to manage equipment stress and panel PID risk.
- A combined approach balancing fixed capacity costs against avoided exceedance charges.



Illustrative representation. Not actual site data.

Outcome Equivalent grid-related charges were reduced to zero during the monitored post-change period following coordinated plant control changes. Revised import capacity requirements were also validated through modelling and data analysis.

What this demonstrates

- Grid compliance and operational data analysis
- Ability to identify root causes rather than symptoms
- Integration of controls, electrical behaviour and contractual requirements
- Quantification of commercial exposure
- Practical recommendations that can be implemented on operational assets

Case Study 2: Voltage and reactive power optimisation on two 6 MW solar sites in Maine

Customer: Utility-scale solar developer and operator

Project: Solar sites in Maine, US, 6 MW export limits

Challenge

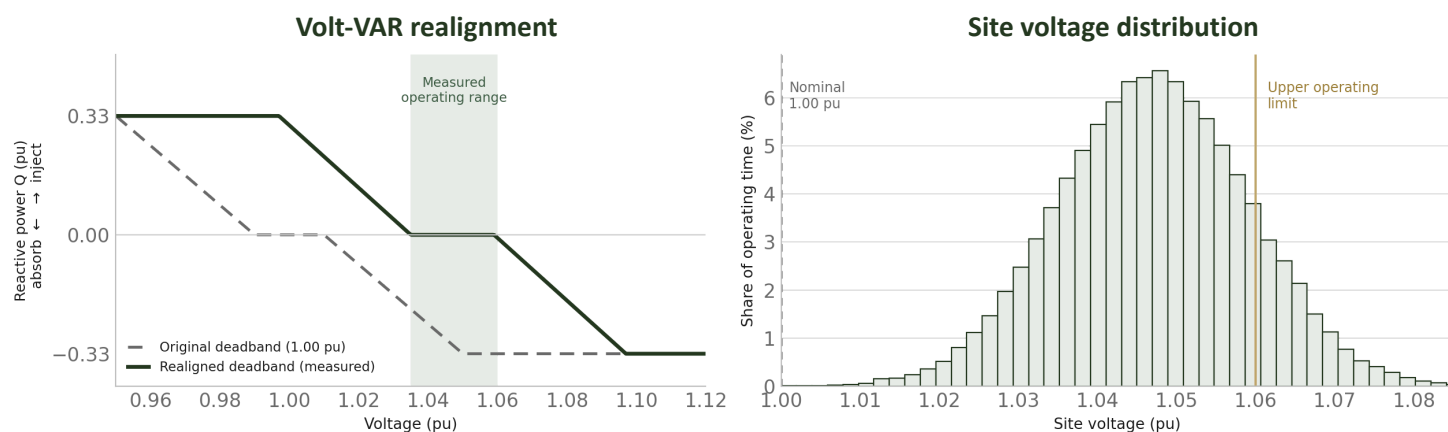
The sites were exporting substantial reactive power due to elevated voltage at the point of interconnection. The Volt-VAR curve loaded into the inverters at commissioning had been set for materially lower voltage conditions than those present in normal local grid conditions. As a result, the plants were not supporting the local network as intended and were operating outside the expected reactive power response.

Dolivo's work

Dolivo produced a diagnostic report assessing point of interconnection voltage statistical distributions, reactive power behaviour, existing Volt-VAR settings, inverter control response, transformer tap configuration and differences between expected and actual operating conditions.

Recommendations

Dolivo proposed a revised Volt-VAR curve aligned with the actual external system conditions, restoring more appropriate and supportive reactive power behaviour.



Illustrative representation. Not actual site data.

Outcome Active power production calculated to increase by approximately 3% through revised control settings aligned with grid conditions.

What this demonstrates

- Voltage and reactive power diagnostics
- SCADA-led control-system analysis
- Translation of operating data and modelling into practical control recommendations
- Identification of performance improvement not visible through standard monitoring alone

Working with Dolivo

An initial engagement typically focuses on a single project or one or more operational assets. From there, Dolivo can work with your teams to extend the approach across a wider portfolio and embed access to specialist power systems expertise more consistently throughout the organisation and project lifecycle.